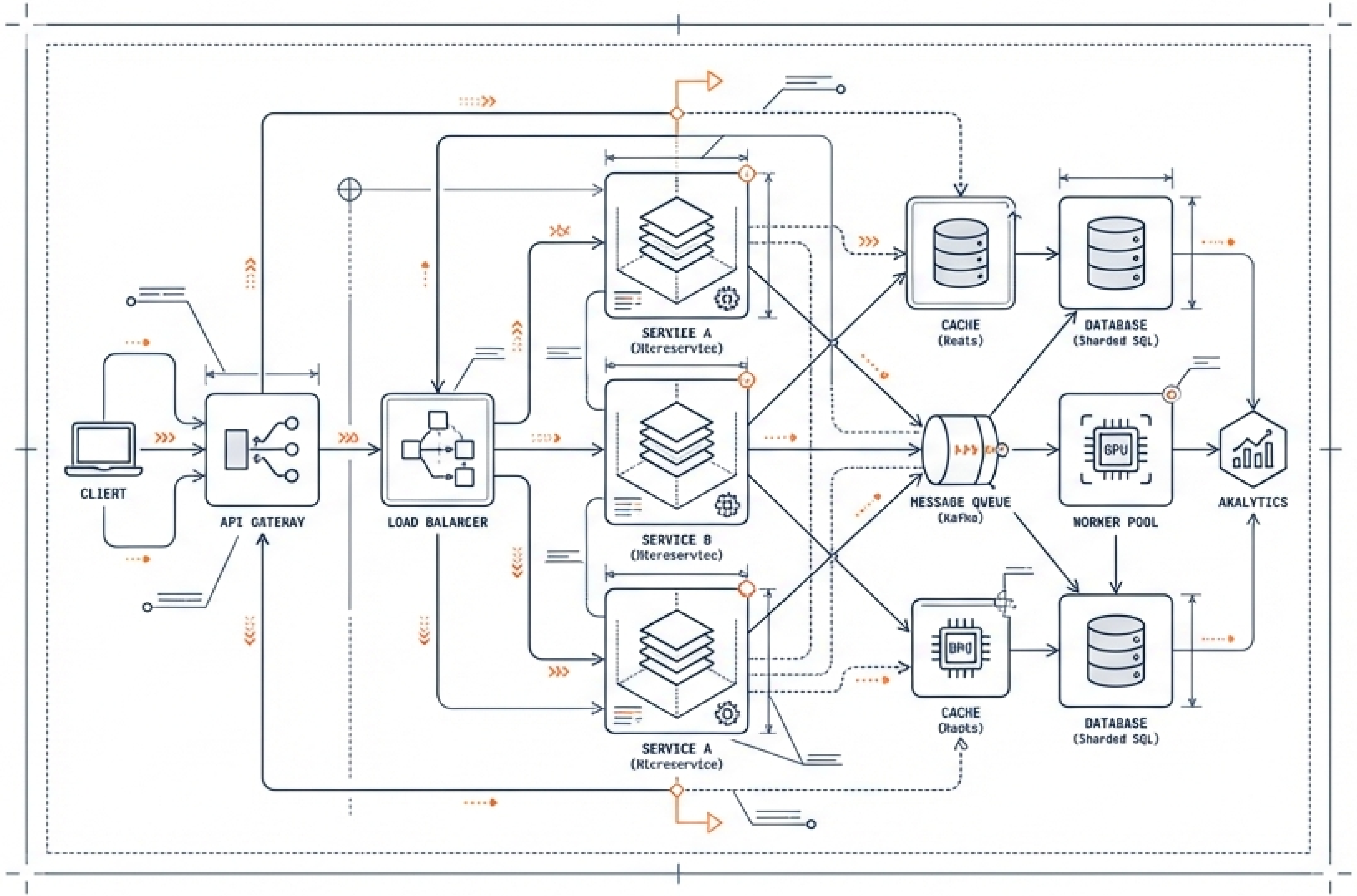
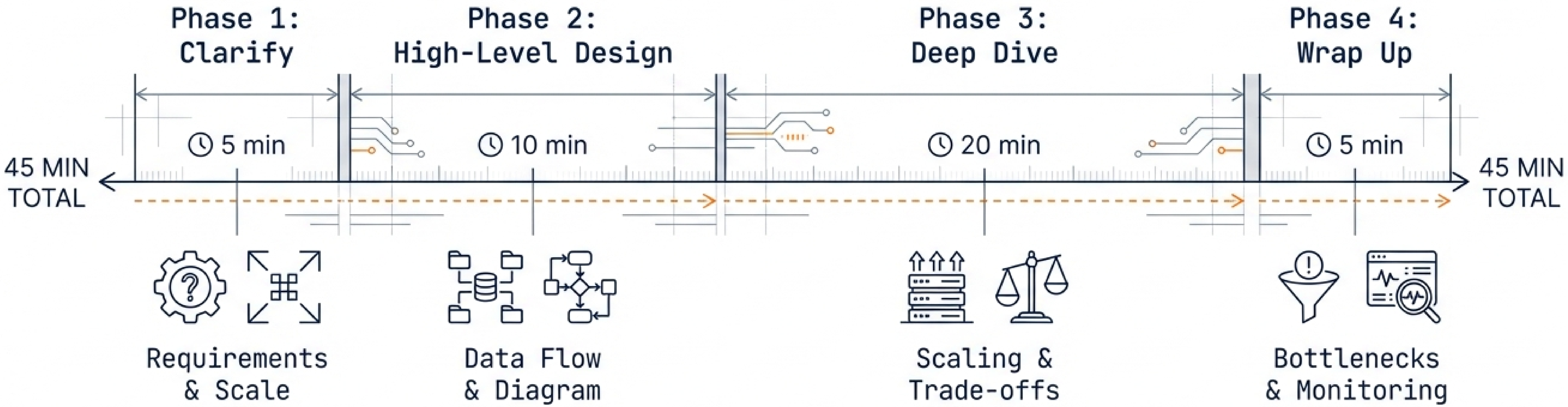
Mastering the System Design Interview

A battle-tested framework to go from ambiguity to scalable architecture in 45 minutes.



The Architect's Blueprint

The 45-Minute Battle Plan



PRO TIP

Never start designing without clarity. A senior engineer asks; a junior engineer assumes.

Back-of-Envelope Estimation Toolkit

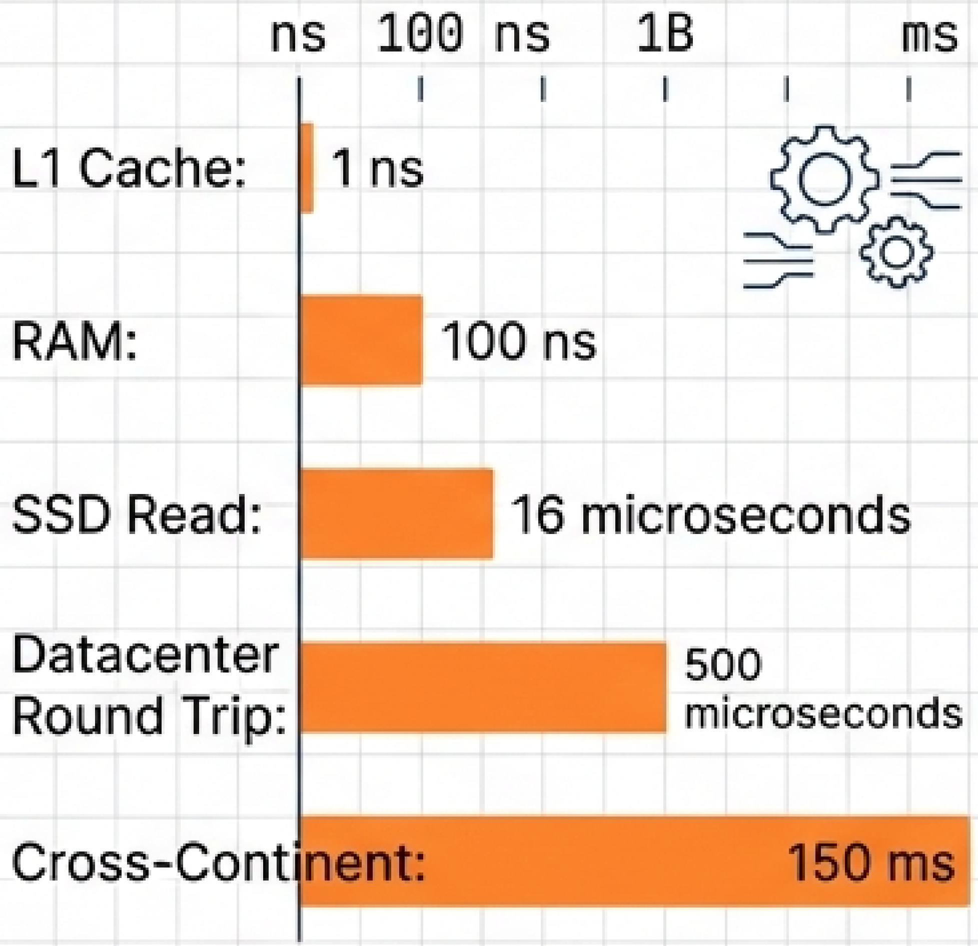
Powers of 10 Rule

1 Day
approx 10^5 sec (100,000)

1 Month
approx 2.5×10^6 sec

1 Year
approx 3×10^7 sec

Latency Numbers (2024)



Capacity Planning

1K QPS:
Single DB

1M QPS:
Sharding Required

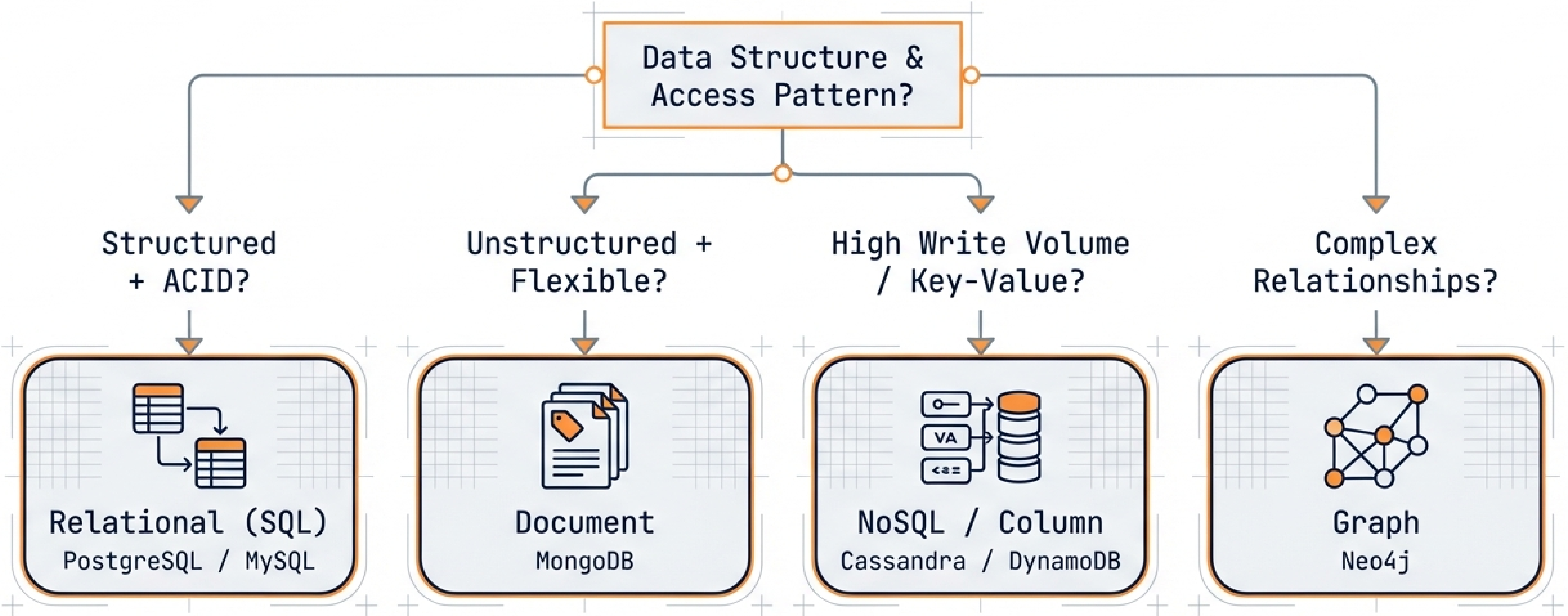
Storage:
1 Char = 1 Byte

NotebookLM

The Core Core Building Blocks

Every large-scale system is composed of these fundamental atoms.

Data Storage Strategy: The Decision Matrix

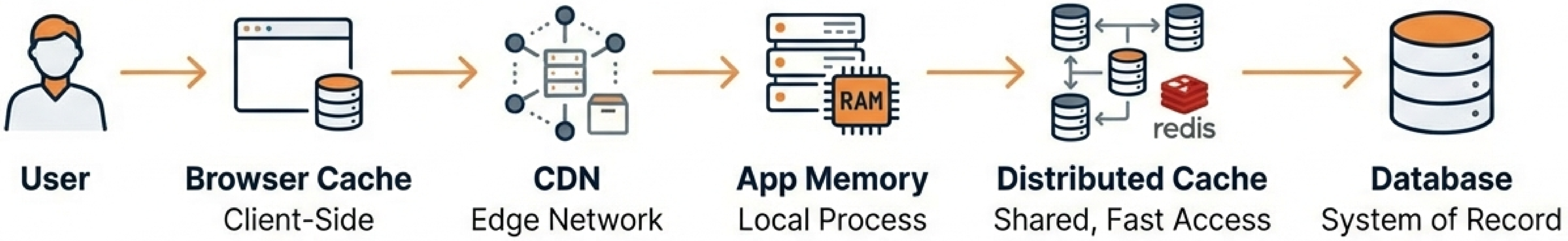


Polyglot Persistence: Use the right tool for the right job.

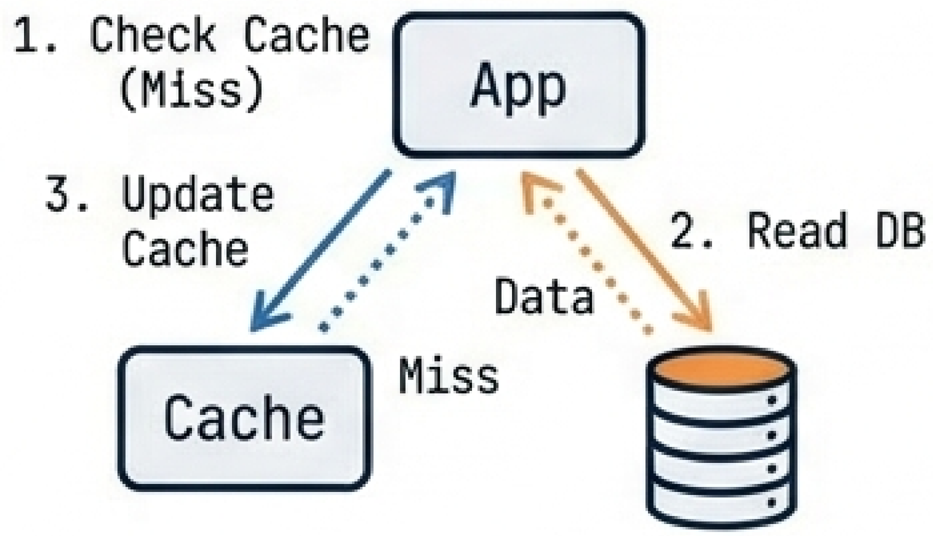
Caching Strategies

The difference between **100ms** and **10ms latency**.

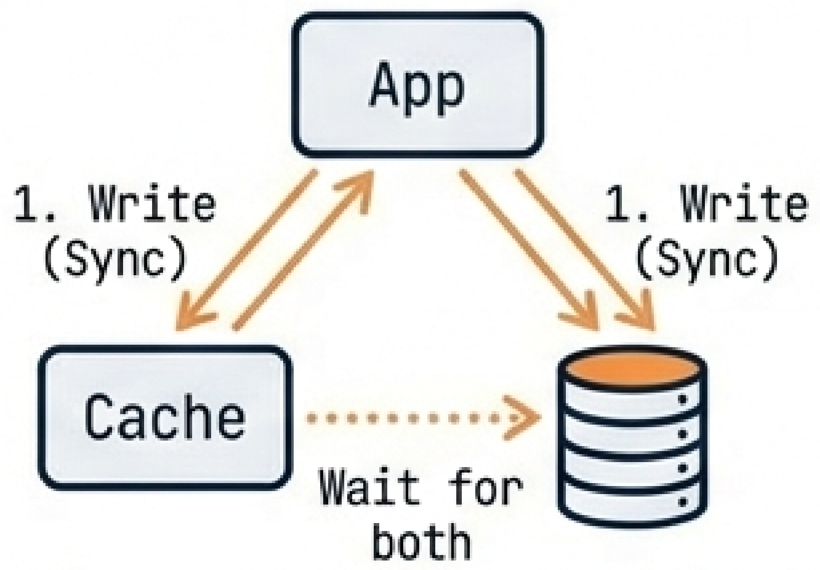
Where to Cache



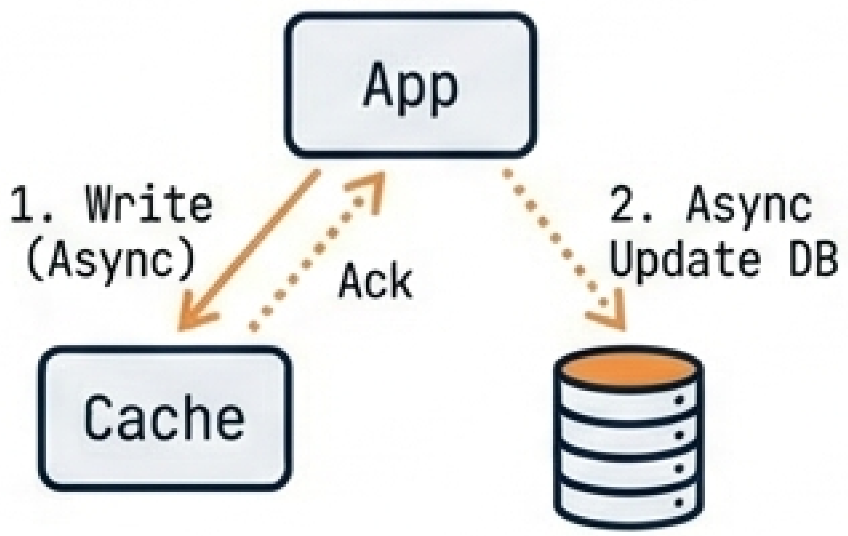
Cache-Aside



Write-Through



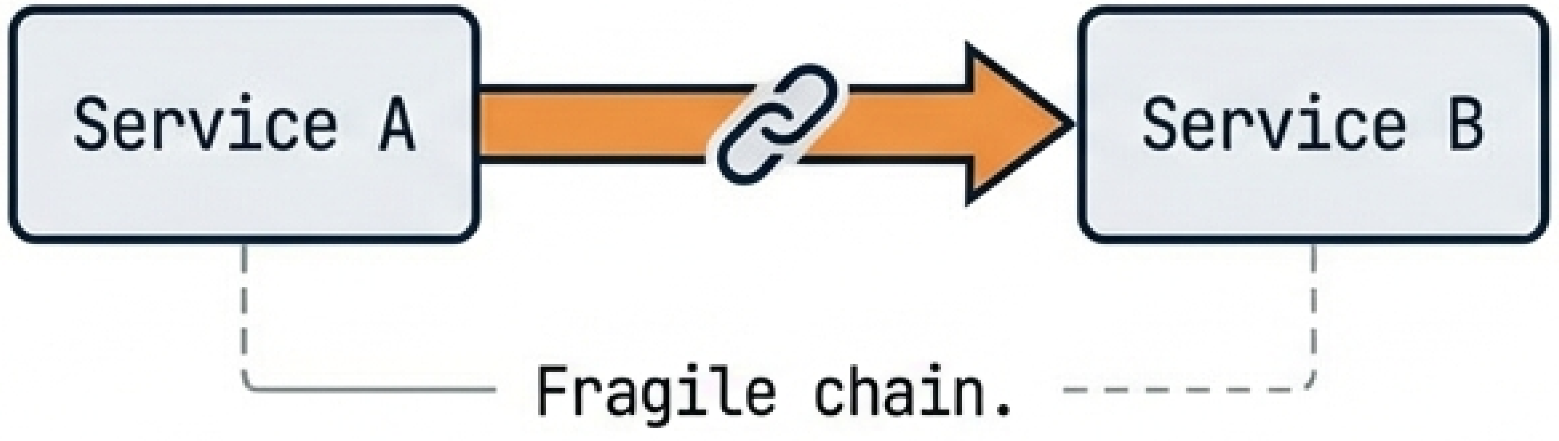
Write-Behind



Eviction Policies: LRU (Least Recently Used) is the standard.

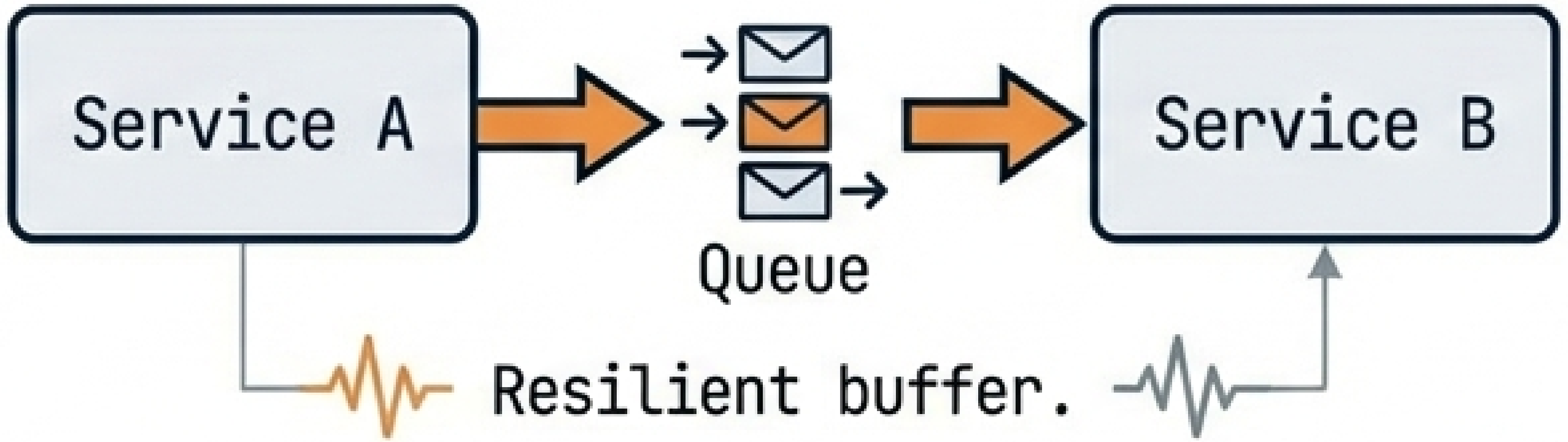
Messaging & Asynchronous Patterns

Synchronous (Tight Coupling)



Queue Type	Best Use Case
Kafka	Stream Processing / Replay
RabbitMQ	Complex Routing
SQS	Simple / Serverless

Asynchronous (Loose Coupling)



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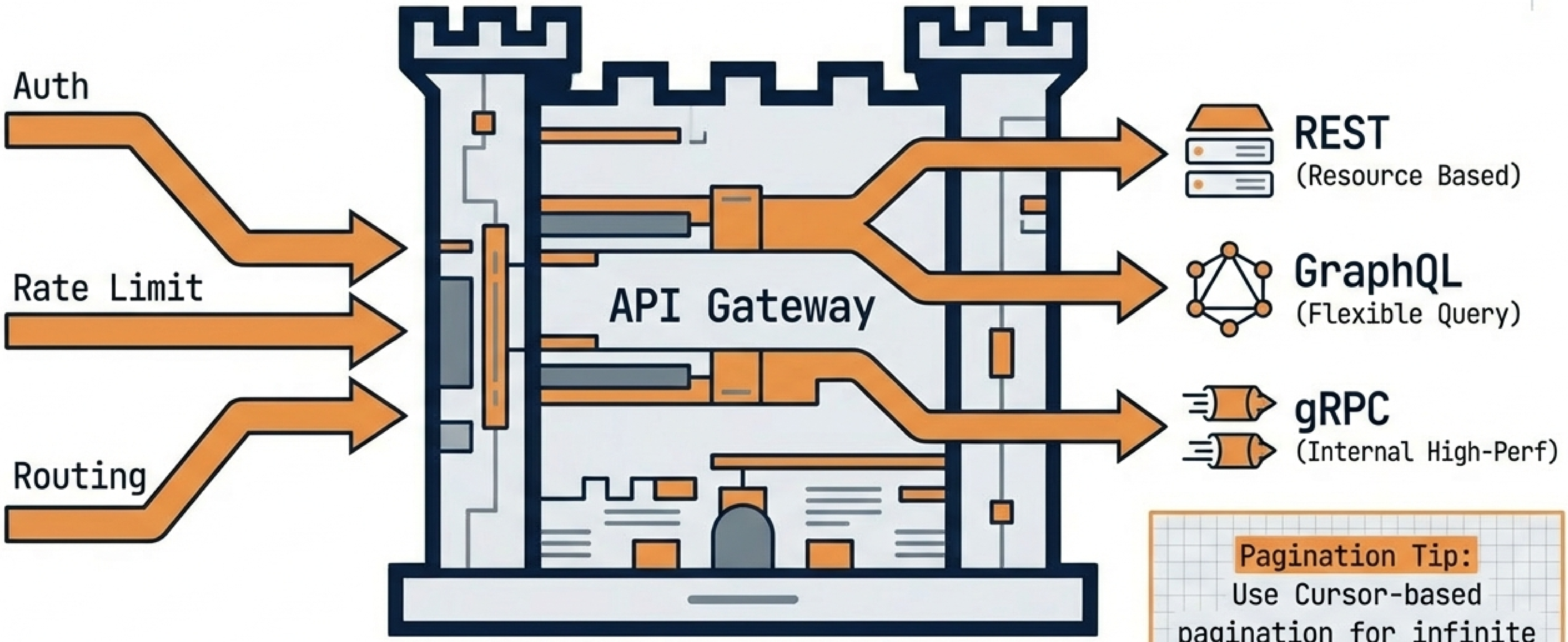
Use Dead Letter Queues (DLQ) to catch failed messages.

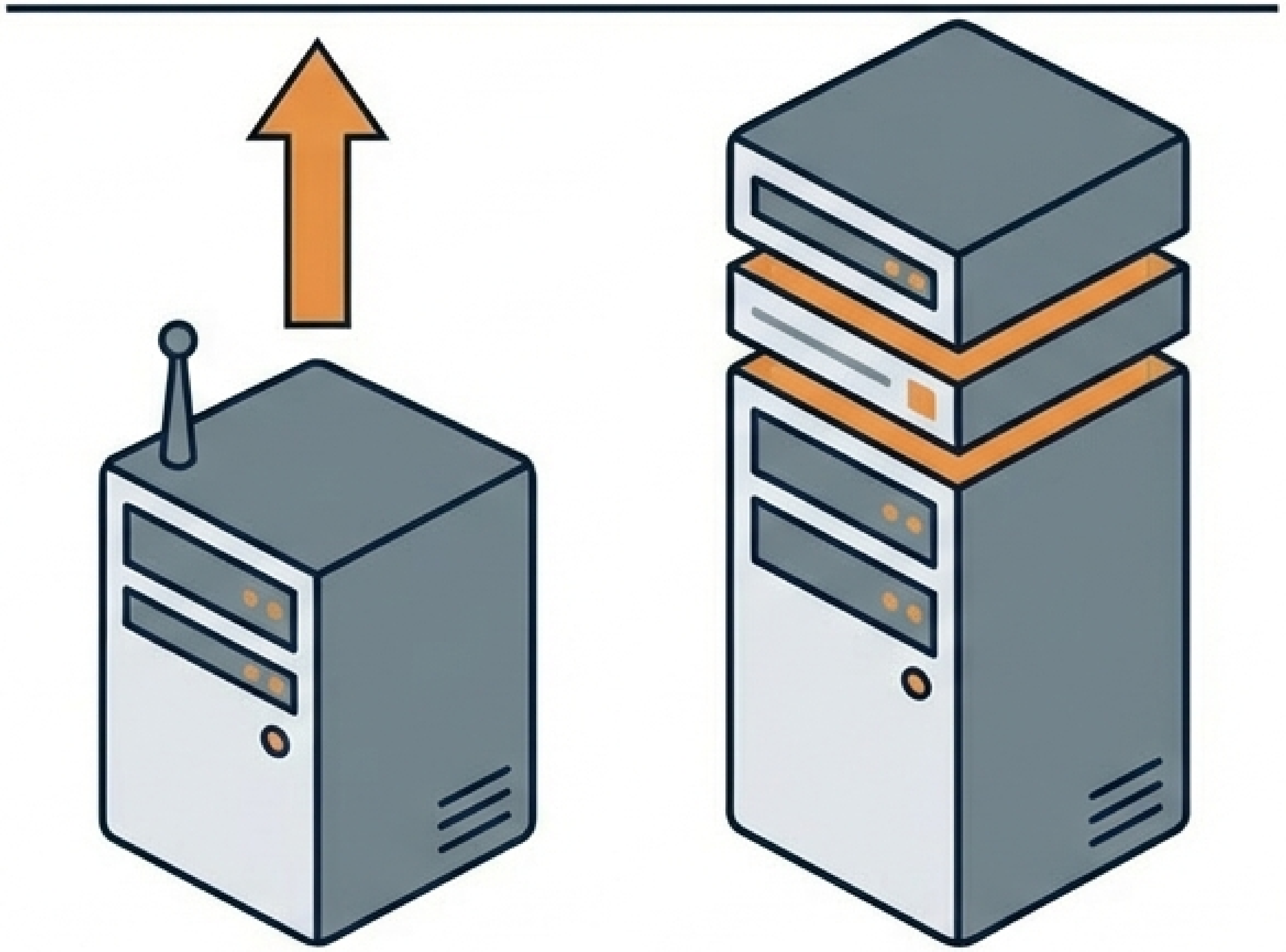
API Design & Gateways



Pagination Tip:
Use Cursor-based pagination for infinite scroll, not Offset.

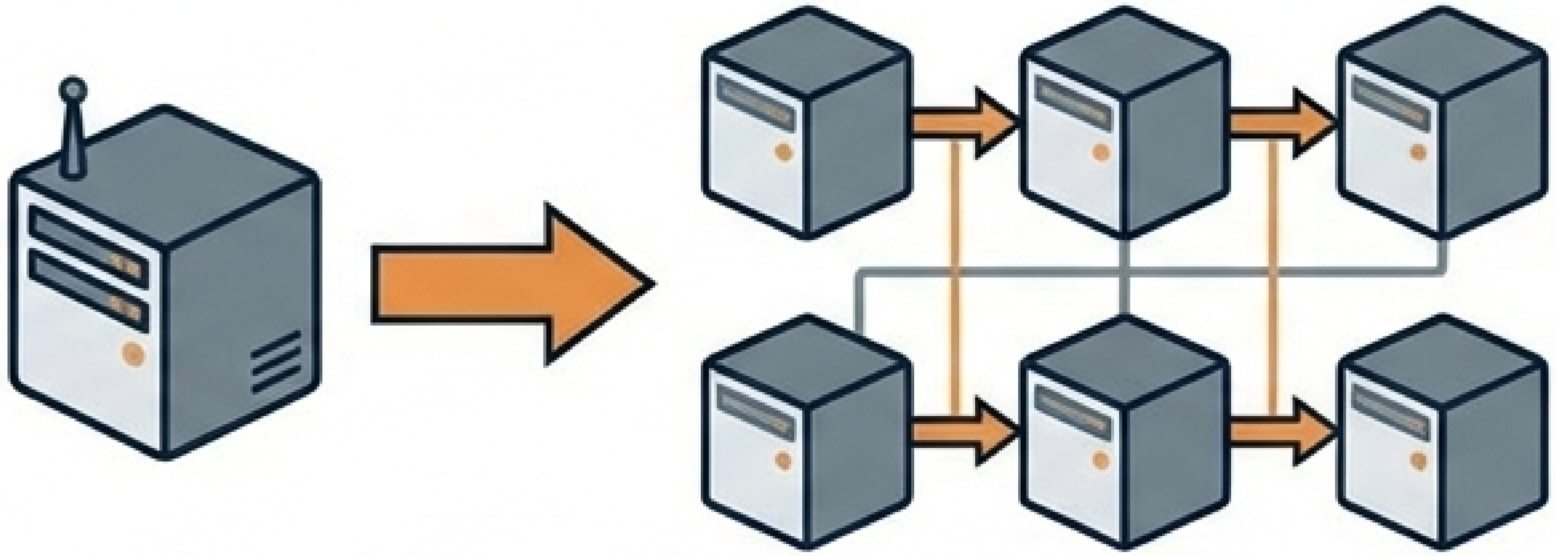
Scalability Patterns

Vertical Scaling



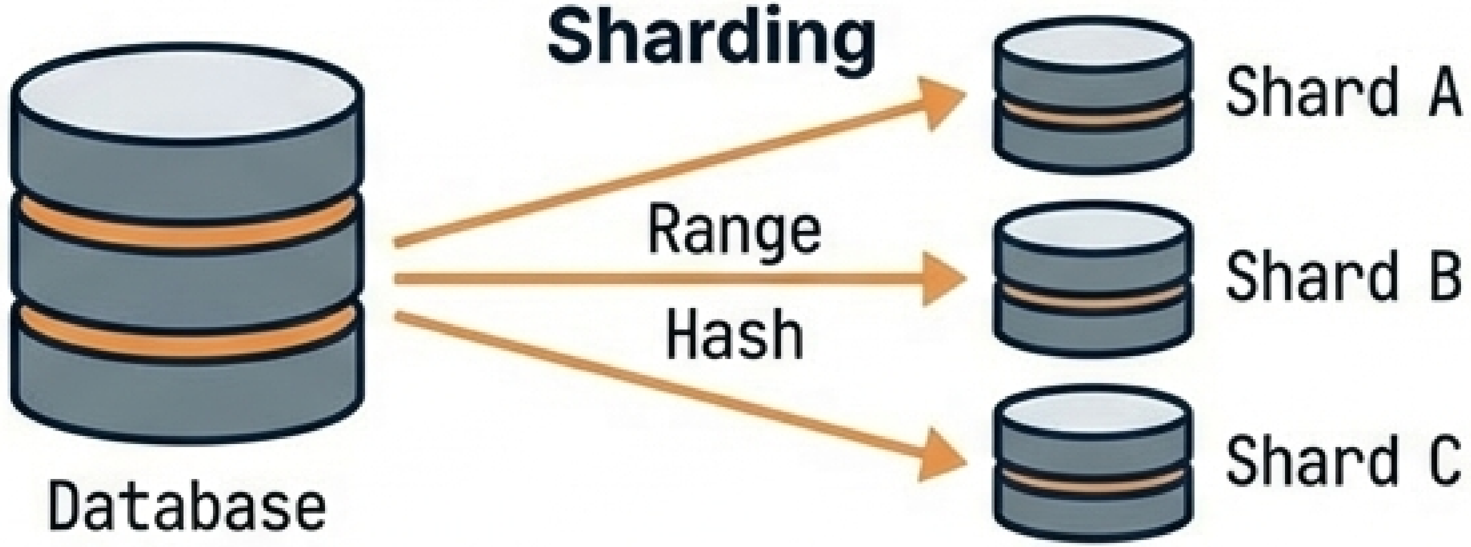
Limited Ceiling / Single Point of Failure

Horizontal Scaling



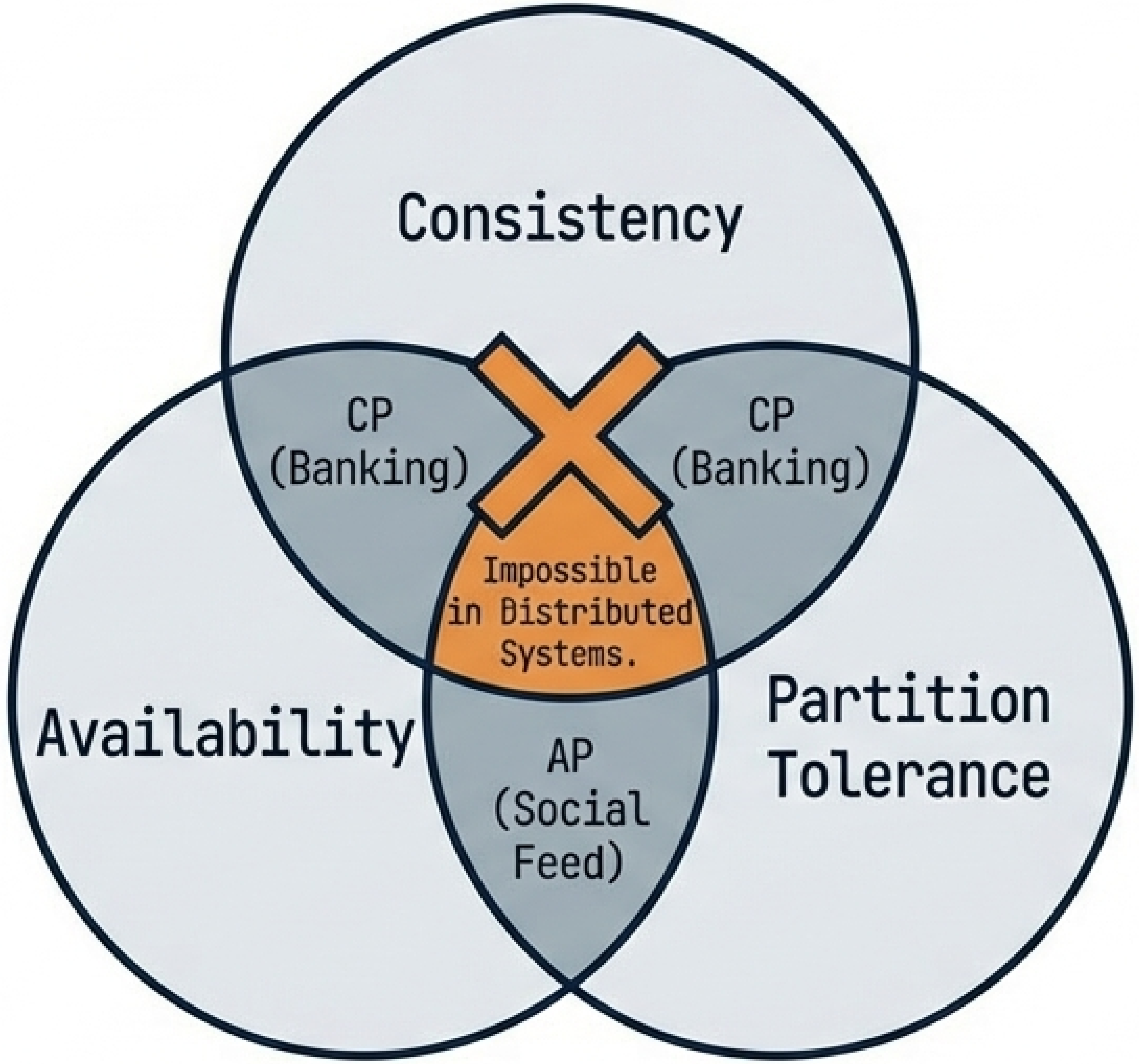
Infinite Scale / High Availability

Sharding



Distributed System Concepts

CAP Theorem



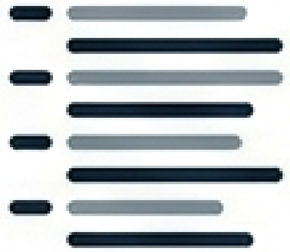
Consistency Models

- Strong (Slow/Correct)
- Eventual (Fast/Stale)

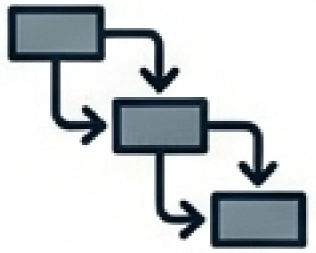
Observability & Reliability

The 3 Pillars of Observability

Metrics
What is happening?
(Prometheus)

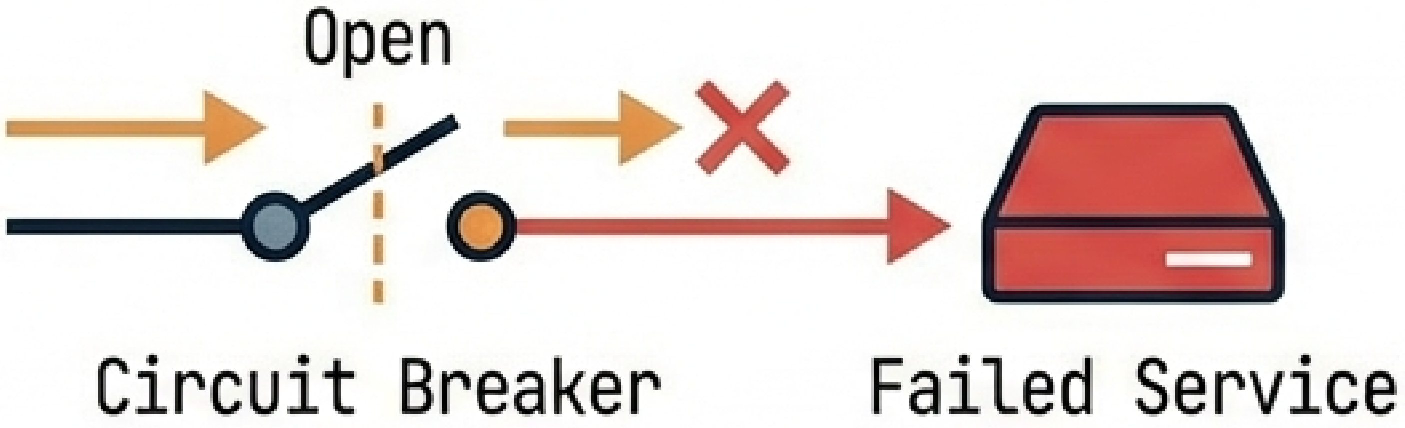


Logs
Why did it happen?
(ELK Stack)



Traces
Where did it happen?
(Jaeger)

Reliability Patterns

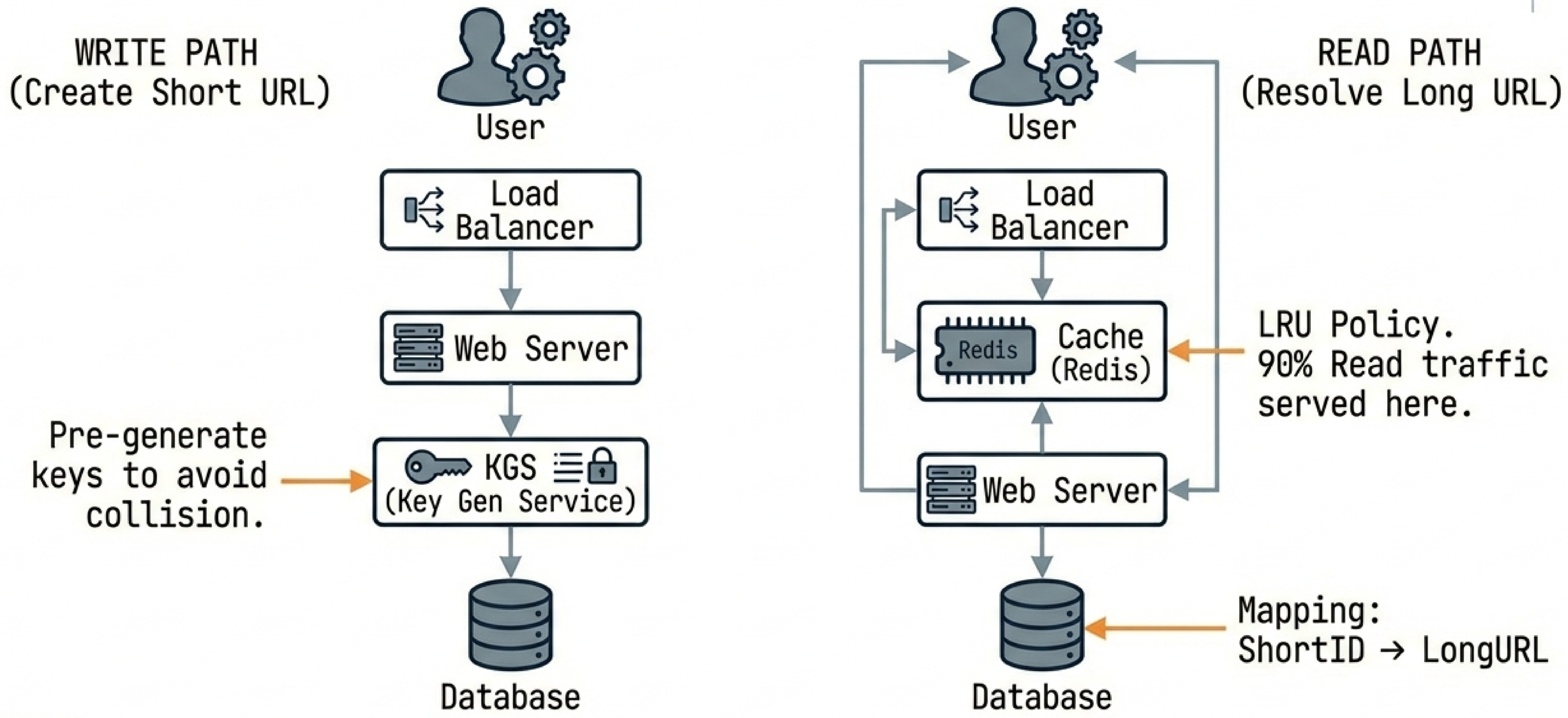


Availability Calculation:
99.9% = 43 minutes downtime/month.

The ‘Big 10’ Common Problems

1. URL Shortener	Challenge: ID Generation (Base62)
2. Rate Limiter	Challenge: Distributed Counting (Redis)
3. Chat System	Challenge: WebSockets & Message Ordering
4. News Feed	Challenge: Fan-out (Push vs Pull)
5. Typeahead	Challenge: Tries & Latency
6. Notification System	Challenge: Queues & Deduplication
7. Ride Sharing	Challenge: Geospatial (QuadTree)
8. Web Crawler	Challenge: Politeness & Duplicates
9. E-commerce	Challenge: Inventory Locking

Architecture Example: URL Shortener



Red Flags vs. Senior Thinking

Red Flags / Junior

- Jumping straight to solution.
- Using one "Magic DB" for everything.
- Ignoring edge cases.
- Silent thinking.

Green Flags / Senior

- Clarify Constraints first.
- Discuss Trade-offs (e.g., Latency vs Consistency).
- Admit Limitations.
- Operational Awareness (Monitoring).

"A senior engineer anticipates failure; a junior engineer hopes for the best."

The Ultimate Cheatsheet

Process

Clarify → High-Level → Deep Dive → Wrap Up

Key Numbers

- L1: 1ns.
- RAM: 100ns.
- SSD: 16us.
- Network: 150ms.

Trade-offs

Latency vs Throughput.

CP vs AP.

SQL vs NoSQL.

Protocols

REST (Standard).

GraphQL (Flexible).

gRPC (Fast).

Scan for System Design Primer